Spotify TALENT RADAR

Hit Song Prediction & Emerging Artists Recognition for Strategic Investment

Predict Here

Spotify Talent Radar -

Upgrade Recommendations

Analyst

BUSINESS QUESTION

Can We Predict Which Songs Will Feature in the Billboard Top-10 within 8 weeks and Identify the Emerging Artists with Hit Potential?

Why It Matters Now:

- Optimise Playlist Strategy
- Early Discovery of Emerging Artists
- Maximize Marketing Return on Investment

Goal:

Deliver a Smart Real-Time Dashboard for Hits Forecasting & Uncovering Emerging Talent

Spotify Talent Radar - What do we Predict ?

Datasets

Spotify Songs

Top 100 Song Rankings Over Time

Song level Features

Key Features

- First - week Song Position
- Artist History (Prior Billboard Entries, Months Since First Seen)
- Artist Collaboration
- 9 Audio Features

Challenge:

Data Scaling

29.5k

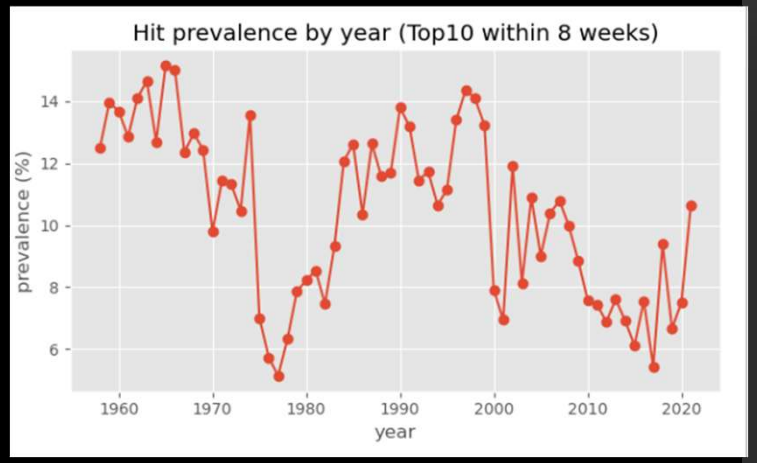
Total Records of first_week Song Position

Imbalanced Dataset

8-12%

Base Hit Rate

Target Variable:

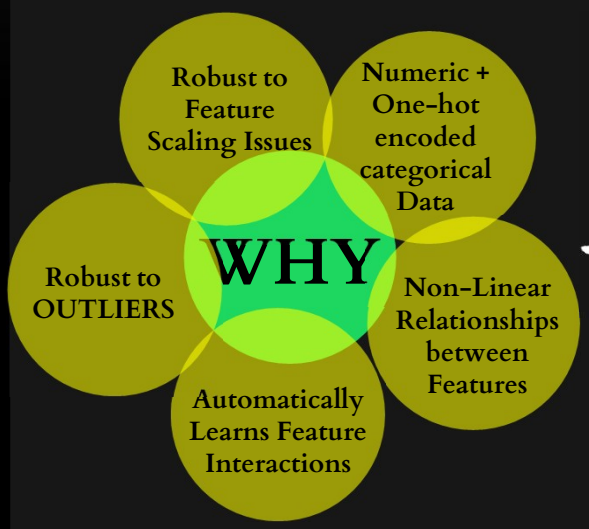


Decline in the speed at which how quickly new releases achieve hit rate

Spotify Talent Radar - Methodology

Our Approach to Build the Confidence

HistGradientBoosting



Ensembled Model

After 400 Iterations

Calibration

Isotonic Regression

Ensures the Model Trusts its own Predictions

Feature Importance

Shapely Additive exPlanations

Top 1200
Highest-Scoring Songs

Top 10
Features Extracted

Spotify Talent Radar - Model Results

Performance Metrics

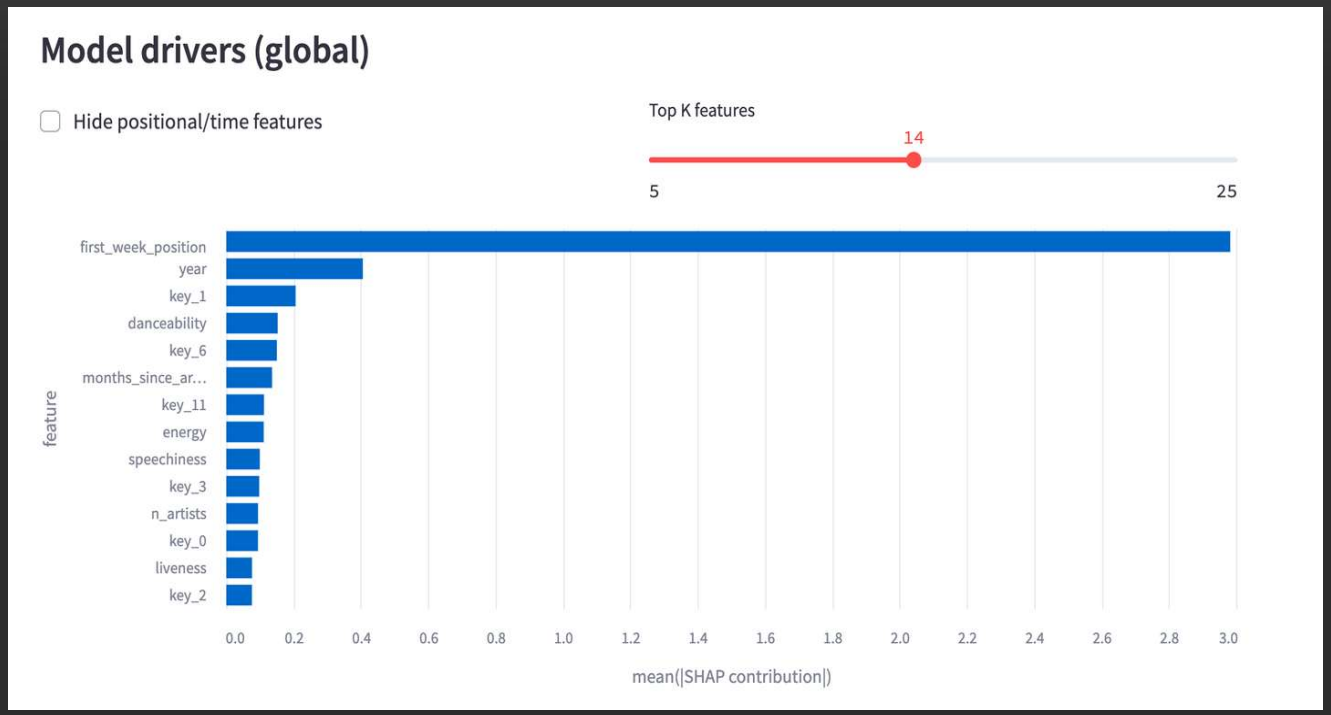
	Evaluation Metric	Value	Emerging Artists
1	PR - AUC	0.631	0.491
2	Precision@20	1.000	1.000
3	Precision@50	1.000	1.000

Interpretation:

The Overall model performs well on unseen data

The Top 20 & 50 Predictions for Overall data and Emerging Artists Subset data are perfectly Accurate.

Key Influential Features in Predicting Hit Song



Spotify Talent Radar

Refresh data

Dataset

Choose file

predictions with flags

Loaded: /Users/priyankayadav/hit-predictor/data/exports/predictions_with_flags.csv

Filters

First week (year range)

2000 2021

1958 2021

Emerging artists only

Search song/artist

taylor swift

Flagging

Mode

Keep existing (if any)

Show top K

Deploy

Spotify Hit Predictor Dashboard

Songs (filtered)	Prevalence	Precision (flagged)	Recall (flagged)
8,892	9.11%	59.81%	39.88%
		↑ Lift ×6.57	

Threshold: 0.3200

Model card & data sources

Business KPIs

Segment	Prevalence	Precision (flagged)	Recall (flagged)	Lift
Overall	9.11%	59.81%	39.88%	656.63%
Emerging	7.74%	53.44%	31.27%	690.41%

Flagged: 540 True hits in flagged: 323

Model card & data sources

Distributions

Spotify Talent Radar -
Business Impact & Recommendations

Impact of Our Model on Real Business Values Across Teams

A & R
Early Identification of High Potential Tracks

Editorial
Earlier Playlist Recommendations

Marketing
5 - 7X Lower Cost Per True Hit Promotion Vs Untargeted

Strategic RoadMap

01 *Integrate News and Social Sentiment Analytics*

02 *Activate Adaptive Thresholds*

03 *Tailor Models With Segments & Track Performance*

Thank You

20k+

Followers

Achieved

Emerging Artist

Despacito

Luis Fonsi, Daddy Yankee

0:23 -3:25